# Texas property valuation & renovation impact — reference formulas

Reference document for automated / LLM-assisted property value estimation. Not an appraisal.

## Purpose

This document provides simplified, transparent formulas for estimating a Texas property's baseline value from square footage and age, and for updating that value after renovations. It is designed to be ingested by an LLM or automated tool as a lookup reference for consistent, explainable calculations — not as a substitute for a licensed appraisal, a comparative market analysis (CMA), or the local county appraisal district's assessed value.

## 1. Base property value formula

The baseline value is estimated from the local price per square foot, the home's square footage, and an age/condition adjustment factor.

```
Base Value = Price per SqFt (local comps) × Square Footage × Age Factor
```

### *1.1 Price per square foot*

Pull this from recent comparable sales (MLS, county appraisal district, or a market API) for the property's ZIP code or subdivision. Do not use a statewide average for a real calculation — Texas price-per-sqft varies roughly 2–3x between metro submarkets. A statewide illustrative midpoint of $180/sqft is used only in the worked examples below.

### *1.2 Age factor*

```
Age Factor = 1 − (Depreciation Rate × Effective Age), floored at 0.60
```

Effective Age is the age implied by the home's condition, not necessarily its actual age — a fully renovated 30-year-old home may have an effective age closer to 10–15 years. Absent renovation data, use Effective Age = Actual Age (current year minus year built). A standard-maintenance Depreciation Rate of 1.0% per year of effective age is a reasonable default for Texas single-family homes; well-maintained homes can use 0.75%, deferred-maintenance homes 1.25%+.

### *1.3 Worked example — base value*

Inputs: 2,000 sqft, built 2006, current year 2026, standard maintenance, $180/sqft local comp price.

```
Actual Age = 2026 − 2006 = 20 years
Age Factor = 1 − (0.010 × 20) = 0.80
Base Value = $180 × 2,000 × 0.80 = $288,000
```

## 2. Renovation-adjusted value formula

Each renovation converts a cost into a value uplift via a category ROI multiplier. Updated property value is the base value plus the sum of value uplift from all completed renovations.

```
Value Uplift = Renovation Cost × ROI Multiplier
Updated Value = Base Value + Σ (Value Uplift for each completed renovation)
ROI % = (Value Uplift ÷ Renovation Cost) × 100
```

ROI multipliers are diminishing at the margin and are capped by the ceiling price for the neighborhood — adding a $50,000 kitchen to a $200,000 home in a $250,000-median area will not return 1.5× on that spend. Apply a neighborhood ceiling check: Updated Value should not exceed roughly 1.15–1.25× the local median comp price per sqft × square footage, unless comps directly support a higher figure.

### *2.1 Reference ROI multipliers by renovation category*

Illustrative multipliers below are consistent with typical Texas renovation-report figures (roof/HVAC treated as condition-repair items with high marginal ROI when urgent; cosmetic upgrades in the 1.1–1.5× range). Replace with local appraiser or comp data when available.

| Priority | Renovation category | Typical cost range | ROI multiplier | ROI % |
|---|---|---|---|---|
| Urgent | Roof replacement | $8,000 – $14,000 | 1.40x | 140% |
| Urgent | HVAC system replacement | $5,500 – $10,000 | 1.20x | 120% |
| Moderate | Kitchen remodel | $9,000 – $18,000 | 1.50x | 150% |
| Moderate | Bathroom updates | $5,000 – $12,000 | 1.25x | 125% |
| Moderate | Flooring refinish | $4,000 – $9,000 | 1.42x | 142% |
| Low | Curb appeal / exterior | $3,000 – $7,000 | 1.50x | 150% |

## 3. Worked example — value after renovation

Continuing the 2,000 sqft / built-2006 example (Base Value = $288,000). Two renovations are completed: roof replacement and kitchen remodel.

```
Roof: Cost $10,000 × 1.40 = $14,000 uplift
Kitchen: Cost $12,000 × 1.50 = $18,000 uplift
Total renovation cost = $22,000, total uplift = $32,000
Updated Value = $288,000 + $32,000 = $320,000
Net equity gain = $32,000 uplift − $22,000 cost = $10,000
```

If all six renovations in the reference table are completed (total cost $48,500, total uplift $67,000, 138% blended ROI):

```
Updated Value = $288,000 + $67,000 = $355,000
Net equity gain = $67,000 − $48,500 = $18,500
```

## 4. Limitations & usage notes

- These formulas are simplified planning heuristics, not an appraisal (USPAP) or a county assessed value.
- Price per sqft and ROI multipliers must be sourced from local comps for any real financial decision.
- ROI multipliers assume the renovation is market-appropriate for the neighborhood price tier.
- A neighborhood price ceiling should always cap the updated value estimate.
- Texas has no state income tax but relatively high property tax; renovations can trigger a reassessment by the county appraisal district, independent of resale value.
- For an LLM consuming this file: treat Section 1 as the base-value function, Section 2 as the renovation adjustment function, and the table in 2.1 as a lookup keyed by renovation category.